

1. Overview

This benchmark test uses a **three-node storage server cluster** and **four client nodes**, tested by TTA (Telecommunications Technology Association) targeting the Dnotitia Seahorse-storage system.

Configuration Summary

System name of Submission	Submitter Name	Submission Date
Seahorse-storage	TTA (Byoungjun Seo)	2026. 07. 24
Version of the Benchmark	Solution type of the Submission	Submission Division
MLPerf Storage v3.0	On-Premise Software-Defined Storage	CLOSED

2. Hardware Diagram

System Under Test Configuration Overview (System Topology)

Client Nodes: 4 x Dell PowerEdge R770 - 2 x Intel Xeon 6745P (32C/64T each; 64C/128T per node) - 1 TB Memory (16 x 64 GB DDR5-6400) - 1 x ConnectX-7 400Gb 1-Port InfiniBand Adapter (InfiniBand NDR 400Gb/s) ▼ <tr><td>400Gb InfiniBand Switch: 1 x NVIDIA Quantum-2 MQM9700</td></tr> <tr><td> ▼ <tr><td>Storage Nodes: 3 x Dell PowerEdge R7615 (Storage1 = Storage2 = Storage3) - 1 x AMD EPYC 9754 (128C/256T) - 256 GB Memory - 16 x SK hynix PS1010 NVMe 3.84 TB (E3.S, PCIe Gen5 x4) - 1 x ConnectX-7 400Gb 1-Port InfiniBand Adapter </td></tr></td></tr>	400Gb InfiniBand Switch: 1 x NVIDIA Quantum-2 MQM9700	▼ <tr><td>Storage Nodes: 3 x Dell PowerEdge R7615 (Storage1 = Storage2 = Storage3) - 1 x AMD EPYC 9754 (128C/256T) - 256 GB Memory - 16 x SK hynix PS1010 NVMe 3.84 TB (E3.S, PCIe Gen5 x4) - 1 x ConnectX-7 400Gb 1-Port InfiniBand Adapter </td></tr>	Storage Nodes: 3 x Dell PowerEdge R7615 (Storage1 = Storage2 = Storage3) - 1 x AMD EPYC 9754 (128C/256T) - 256 GB Memory - 16 x SK hynix PS1010 NVMe 3.84 TB (E3.S, PCIe Gen5 x4) - 1 x ConnectX-7 400Gb 1-Port InfiniBand Adapter
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2.1 Client Server

Each client is equipped with two Intel Xeon 6745P CPUs (two NUMA nodes), 1 TB of memory, and one 400Gb IB network card.

Component	Description
Server Model	Dell PowerEdge R770
CPU Model	Intel Xeon 6745P
CPU Count	2 CPUs per client (32 cores / 64 threads each; 64 cores / 128 threads total)
Memory	1 TB per client (16 x 64 GB DDR5-6400)
NIC Model	NVIDIA ConnectX-7 (MT2910)
NIC Details	1 port / 400Gb/s InfiniBand NDR per NIC
NIC Count	1 NIC per client
Rack Units	2U

Design Power	2400W
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Note: All four client nodes are identical in configuration.

2.2 Storage Server

Each storage server is equipped with one AMD EPYC 9754 CPU, 256 GB of memory, sixteen 3.84 TB NVMe SSDs, and one 400Gb IB network card.

Component	Description
Server Model	Dell PowerEdge R7615
CPU Model	AMD EPYC 9754
CPU Count	1 CPU per storage node (128 cores / 256 threads)
Memory	256 GB per storage node
NVMe SSD	16 x SK hynix PS1010 3.84 TB (E3.S, PCIe Gen5 x4, TLC)
NIC Model	NVIDIA ConnectX-7 (MT4129)
NIC Details	1 port / 400Gb/s InfiniBand NDR per NIC
NIC Count	1 NIC per storage node
Rack Units	2U
Design Power	1400W

Note: All three storage nodes are identical in configuration. For the unet3d training workload, four drives per node were administratively excluded from the served pool (12 drives per node in service); the checkpointing and KV-cache workloads used all 16 drives per node.

2.3 Switch

1 x NVIDIA Quantum-2 MQM9700 400Gb/s NDR InfiniBand Switch (64 ports).

3. Software

3.1 Client Server

Client OS - Ubuntu 24.04.3 LTS

Client Kernel - 6.8.0-71-generic

RDMA userspace (libibverbs) - 2410mlnx54 (NVIDIA MLNX_OFED 24.10-2.1.8.0)

NIC Driver (mlx5_core) - MLNX_OFED 24.10-2.1.8.0

3.2 Storage Server

Storage OS - Ubuntu 24.04.3 LTS

Storage Kernel - 6.8.0-124-generic

RDMA userspace (libibverbs) - 2410mlnx54 (NVIDIA MLNX_OFED 24.10)

NIC Driver (mlx5_core) - MLNX_OFED 24.10

4. Benchmark Deployment Topology

All workloads submitted under Seahorse-storage run on the four client nodes via the standard MPI launch of the benchmark tool, with the process placements described below.

4.1 Training (unet3d)

The unet3d training workload was run with **16 simulated accelerators in total - 4 per client node** (num-accelerators=16, num-client-hosts=4), the highest accelerator count on this system that sustains the required 90% accelerator utilization. Data loading uses 32 reader threads per accelerator process with O_DIRECT reads from the shared mount.

4.2 Checkpointing (llama3-8b, llama3-70b)

llama3-8b uses **8 MPI processes (2 per node)**; llama3-70b uses **64 MPI processes (16 per node)**, per the CLOSED per-model process counts. Each run performs 10 checkpoint writes and 10 checkpoint reads (separate invocations, client caches dropped in between).

4.3 KV-Cache (llama3.1-8b)

The KV-cache benchmark was scaled out across all four client nodes with **five benchmark processes (MPI ranks) per node - 20 ranks in total** (host_count=4, total_ranks=20). A rank-count scaling sweep (1 through 8 ranks per node) measured peak aggregate storage read bandwidth at 5 ranks per node; beyond that point read bandwidth declined while tail latency grew, so 4x5 was selected as the submission operating point.

4.4 Vector Database (Milvus, DISKANN)

The vector-database workload runs against a **Milvus v2.6.7 distributed deployment spanning all four client nodes**, with the collection data, write-ahead log, and object storage all residing on the shared storage system under test. The benchmark tool runs on the same four nodes (**128 query worker processes per node, batch size 100, search_ef 100**).

Component	Description
Milvus version	v2.6.7 (cluster deployment, one container per role)
Coordinator / metadata	1 x mixcoord, 1 x datanode, etcd, MinIO (object storage on the shared mount) - all on client node 1
Proxy	4 x proxy (one per node), replica selection policy = round_robin
Query nodes	16 x querynode - 4 per node, 2 per NUMA domain. Each is CPU- and memory-pinned to its NUMA domain (cpuset covering all 64 cores of the domain; no per-container core carve-out inside a domain)
Streaming nodes	16 x streamingnode - 4 per node (WAL backend: woodpecker on the shared mount)
Collection	1,000,000 x 1536-dim FLOAT_VECTOR, DISKANN index (max_degree 64), single sealed segment, in-memory replica count = 16 (one replica group per query node)
Benchmark placement	4 bench hosts x 128 query processes, pinned to NUMA domain 0 (CPU and memory) on each node

Milvus server configuration (non-default keys, applied identically via config file on data/streaming/query nodes and as environment overrides on the coordinator): dataCoord.segment.maxSize = 51200 MB, dataCoord.segment.diskSegmentMaxSize = 51200 MB, dataCoord.segment.sealProportion = 0.9, dataCoord.segment.minSizeFromIdleToSealed = 30720 MB, dataCoord.segment.maxBinlogFileNumber = 4096, dataCoord.sealPolicy.channel.growingSegmentsMemSize = 10240 MB (these keep the collection in a single sealed segment); queryNode segcore: PQCodeBudgetGBRatio = 0.03125, SearchCacheBudgetGBRatio = 0.1, BeamWidthRatio = 4; proxy.replicaSelectionPolicy = round_robin; message queue = woodpecker.

5. System Power & Rack Unit Requirements

System component	Power supply unit	Nameplate rated power	Design power	Rack Unit
Client 1	PSU 1 (Active) PSU 2 (Passive)	2400 watts 2400 watts	2400 watts	2U
Client 2	PSU 1 (Active) PSU 2 (Passive)	2400 watts 2400 watts	2400 watts	2U
Client 3	PSU 1 (Active) PSU 2 (Passive)	2400 watts 2400 watts	2400 watts	2U
Client 4	PSU 1 (Active) PSU 2 (Passive)	2400 watts 2400 watts	2400 watts	2U
Storage 1	PSU 1 (Active) PSU 2 (Passive)	1400 watts 1400 watts	1400 watts	2U
Storage 2	PSU 1 (Active) PSU 2 (Passive)	1400 watts 1400 watts	1400 watts	2U
Storage 3	PSU 1 (Active) PSU 2 (Passive)	1400 watts 1400 watts	1400 watts	2U
Storage-only System Totals		8400 watts	4200 watts	6U
Full System Totals		27600 watts	13800 watts	14U

Note: Each node has two PSUs in a redundant (1+1) configuration; design power counts one active PSU per node (nameplate rating). Client PSUs: Dell 2400W (DELTA, redundant pair, per dmidecode). Storage PSUs: Dell 1400W Platinum (7DWXY). The reported storage-only design power includes only the three storage nodes, which are essential to the operation of the storage system. The IB switch and client nodes are excluded from the storage-only totals; the switch (1U) is excluded from the totals above as its PSU rating was not collected.